Automated Hydroponic System using IoT and Cloud Computing

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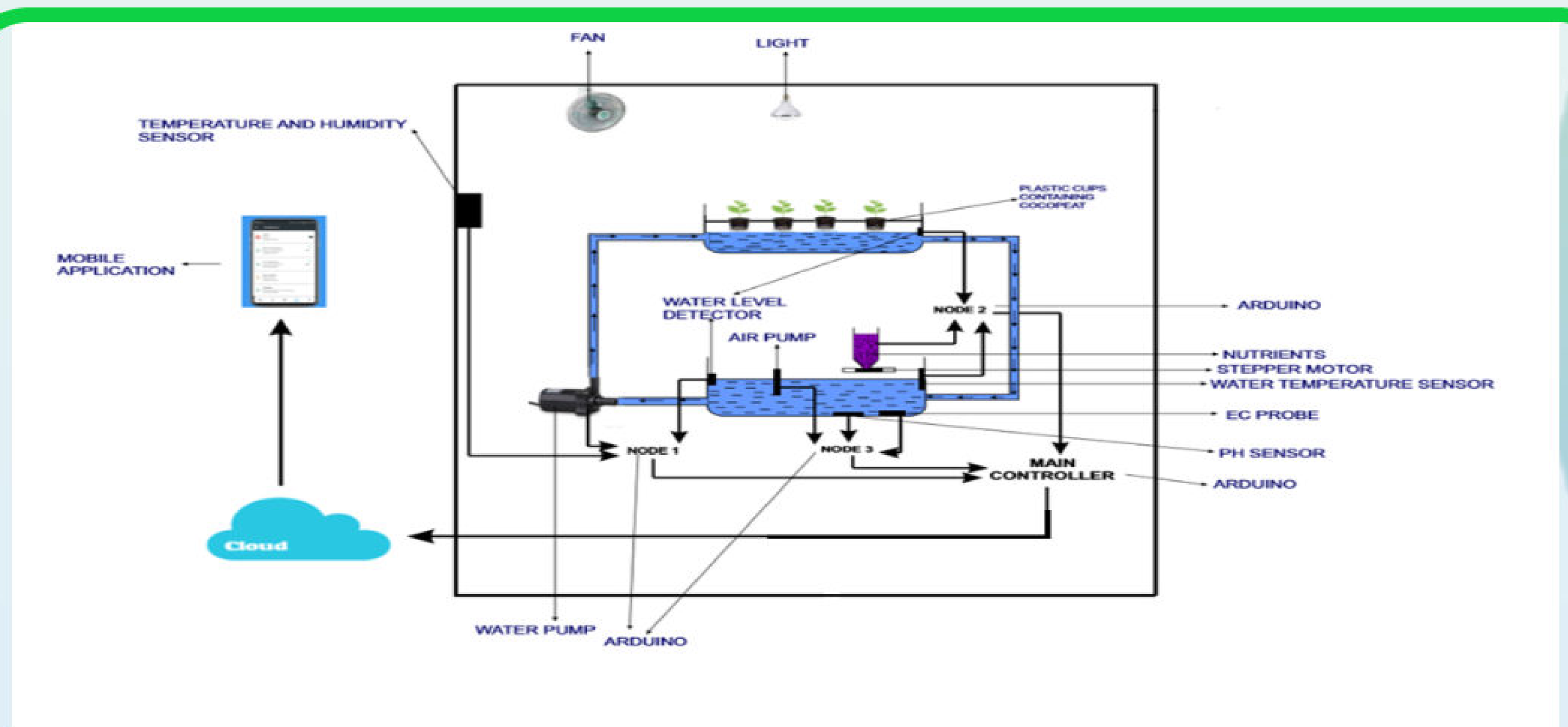
Abstract

Our project explores the idea of automation in regards of hydroponic system. Hydroponics is a method of growing plants in a water based, nutrient rich solution without the use of soil. The purpose is to make it manageable and accessible for the masses so that the people of Bangladesh can re-establish the slowly falling agricultural sector with advanced technology. We are integrating internet of things (IoT) and cloud computing to remove the need for constant maintenance and make it so that the new comers has more access.



Method with system diagram

There are three main parts of our system architecture:

1. Hydroponic System : We used NFT in our system.
2. Automation using IoT and Cloud Computing : We used arduino microcontrollers to control our sensors and wifi module to send the data to 'ThinkSpeak'.
3. Application to control the system : Our app keeps the user constantly updated by fetching the data.

Novelty of Project and Significance

The reason for the scarcity of the use of hydroponics system is because it is extremely sensitive and is on the expensive side even more so if you are trying to buy the whole automated system from another country. Our system solves both the problems. The use of automation makes it highly manageable and starting an industrial production in this country will make it more affordable, accessible and a good investment.

Social and economical impact

- Crops grow 2 times faster, attracts fewer pests and diseases resulting to a little to no use of pesticides, herbicides and fertilizer.
- It can produce any plants viz. foreign, ornamental, food, medicine etc. (except for woody plants)
- Takes less space and is easier to manage.
- The nutrient rich water is reused so there's less wastage and subsequently more cost effective.